

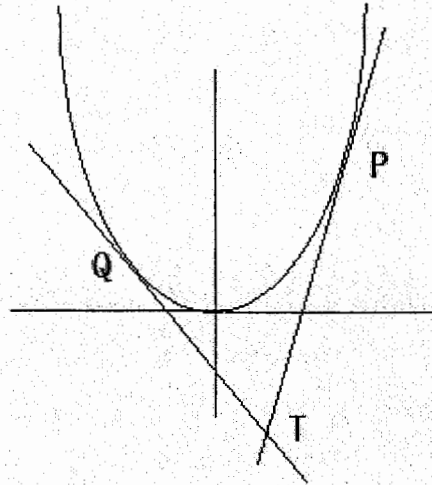
Question 1 (12 Marks) Use a separate piece of paper

Marks

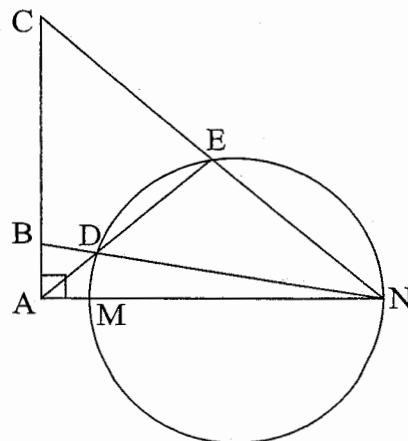
- a) Evaluate $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$ 2
- b) Let A be the point (-8 , -3) and B the point (4 , 7). Find the coordinates of the point that divides AB externally in the ratio 1 : 2. 2
- c) Use the substitution $u = \tan x$ to evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^3 x \sec^2 x dx$ 3
- d) State the domain and range of the function $f(x) = 2 \sin^{-1} \frac{x}{3}$ 2
- e) Solve for x $\frac{3x}{x-1} \leq 2$ 3

Question 2 (12 Marks) Use a separate piece of paper

- a) Find $\frac{d}{dx} x \tan^{-1} x^2$ 3
- b) (i) Write $5 \sin x + 3 \cos x$ in the form $R \sin(x + \alpha)$ where $0 \leq \alpha \leq 90^\circ$ and $R \geq 0$ 2
- (ii) Hence or otherwise solve the equation $5 \sin x + 3 \cos x = 4$ for x to the nearest degree for $0 \leq x \leq 360^\circ$ 2
- c) Find the term independent of x in the expansion of $\left(2x - \frac{1}{x^2}\right)^9$ 3
- d) Evaluate $\int_0^{\frac{\pi}{4}} \sin^2 x dx$ 2



- a) The diagram above shows the parabola $x^2 = 4ay$ the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola.
- (i) Find the equation of the tangent at P . 3
 - (ii) State the equation of the tangent at Q . 1
 - (iii) Find the coordinates of T the point of intersection of the two tangents 2
- b) When $P(x)$ is divided by $(x + 1)$ the remainder is 3, when $P(x)$ is divided by $(x - 2)$ the remainder is -5 . What is the remainder when $P(x)$ is divided by $(x + 1)(x - 2)$. 3
- c) In the figure, M, N, E and D are the points on the circle. MN is a diameter. NE is produced to C and NM is produced to A such that the $CA \perp AN$. AE meets the circle at D . ND is produced to meet CA at B .
- (i) Prove $\triangle MEN$ is similar to $\triangle CAN$ 1
 - (ii) Hence or otherwise show that B, C, E and D are concyclic 2



Question 4 (12 Marks) Use a separate piece of paper

Marks

- a) Assume that the rate at which a body cools in air is proportional to the difference between its temperature T and the constant temperature of the surrounding air A . This can be expressed as

$$\frac{dT}{dt} = -k(T - A) \text{ where } t \text{ is time in minutes and } k \text{ a constant}$$

- (i) Show that $T = A + Ce^{-kt}$ where C is a constant is a solution to the differential equation.

1

- (ii) If molten steel cools from an initial temperature of 500° to 200° in 15 minutes with an air temperature of 20° . Find the values of A , C and k .

3

- (iii) How long does it take, to the nearest minute, for the steel to cool to 100° .

1

- b) Prove by Mathematical induction that

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$$

3

- c) The polynomial $P(x) = x^3 - x^2 - 2x + 5$ has a root between -1 and -2 . Using $x = -1$ as the first value use Newton's Method once to find a better approximation. (Give your answer to two decimal places)

2

- d) Use the definition of the derivative $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

to find $f'(x)$ when $f(x) = \sqrt{x}$

2

Question 5 (12 Marks) Use a separate piece of paper

- a) How many arrangements of the letters of the word WALLABY are possible.

1

- b) For positive integers n and r with $r < n$ show that

$${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$$

$$\text{where } {}^nC_r = \frac{n!}{r!(n-r)!}$$

3

- c) A gambling game consists of three fair dice being rolled and betting on a particular number appearing on one of the uppermost faces. If such a game is played and the chosen number is 6

- (i) What is the probability that no 6's will appear

1

- (ii) What is the probability that at least one 6 will appear

1

- (iii) If five such games are played, using a binomial expansion or otherwise, find the probability that exactly three turns will include at least one 6. (leave answer in index form)

2

- (iv) Find the probability that at least one game will include at least one 6. (leave answer in index form)

2

- d) Show that $\cos^{-1} \frac{1}{\sqrt{5}} + \cos^{-1} \frac{1}{\sqrt{10}} = \frac{3\pi}{4}$

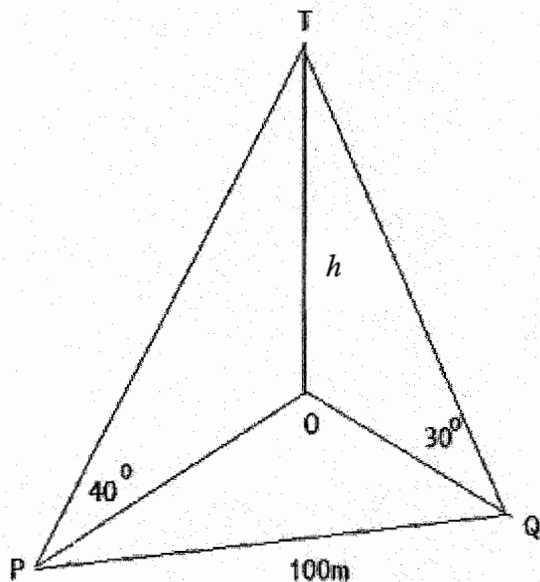
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Question 6(12 marks) Use a separate piece of paper

Marks

- a) By putting $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ in the equation of S.H.M. $\frac{d^2x}{dt^2} = -n^2x$
- (i) Show that $v^2 = n^2(a^2 - x^2)$ where a is the amplitude. 2
- (ii) A particle P performing S.H.M. in a straight line about a point O has speeds of 5m/s and 3m/s at two points A and B which are 0.2 m and 0.6 m respectively from O.
Find the amplitude and frequency of the motion. 2
- (iii) Find the length PO at the instant the velocity v the particle is $\frac{2}{3}$ the maximum velocity of the motion. 2

b)



A surveyor stands at a point P due south of a tower OT of height h , and finds the angle of elevation of the top of the tower to be 40° . And then walks 100m to a point Q, so that the angle POQ is 90° , and finds that the angle of elevation from Q is 30°

- (i) Find expressions for OP and OQ in terms of h . 2
- (ii) Show that $h = \frac{100(\tan 40^\circ \tan 30^\circ)}{\sqrt{\tan^2 40^\circ + \tan^2 30^\circ}}$ 2
- (iii) Find the bearing of P from Q. 2

Question 7 (12 Marks) Use a separate piece of paper

Marks

- a) A coal loader is stacking coal on a flat surface, in the shape of cone.
The cone has a semi vertical angle of 30° . If the coal is being deposited
at the rate of $1\text{ m}^3/\text{min}$, find
- (i) An expression for the volume of the cone in terms of the radius only 2
 - (ii) The rate at which the radius is changing when the radius is 2m. 2
- b) (i) Find the largest positive domain of the function $f(x) = x^2 - 4x + 5$
for which $f(x)$ has an inverse function $f^{-1}(x)$ 1
- (ii) Find $f^{-1}(x)$ and hence sketch the graphs of $f(x)$ and $f^{-1}(x)$
on the same set of axes. 2
- c) A particle is projected with a velocity of $V\text{ m/s}$ at an angle of θ° . Using
- $$x = V \cos \theta t$$
- $$y = V \sin \theta t - \frac{1}{2}gt^2$$
- $$y = V \sin \theta - gt$$
- (There is no need to prove these results)
- (i) Find an expression for the maximum height reached by the projectile 1
 - (ii) Prove that the Cartesian equation of the particle is
- $$y = \tan \theta x - \frac{g \sec^2 \theta x^2}{2V^2} \quad 2$$
- (iii) If the particle passes through a point at height b , and horizontal distance
 a from the origin, prove that the maximum height reached is given by
- $$\frac{1}{4} \left[\frac{a^2 \tan^2 \theta}{a \tan \theta - b} \right] \quad 2$$

Question 1.

(a) $I = \int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$
 $= \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}}$
 $= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0$
 $I = \frac{\pi}{3}$ (2m)

(b) $A(-8, -3) \quad B(4, 7) \quad 1:-2$
 $P_x = \frac{-2(-8) + 1 \times 4}{-1} \quad P_y = \frac{-2(-3) + 1 \times 7}{-1}$

$P(-20, -13)$ (2m)

(c) $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^3 x \sec^2 x \, dx$

let $u = \tan x$
 $\frac{du}{dx} = \sec^2 x$

$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} u^3 \, du$
 $= \left[\frac{u^4}{4} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$

$= \left[\frac{\tan^4 x}{4} \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$

$= \frac{9}{4} - \frac{1}{4}$

$I = 2$ (3m)

(d) $f(x) = 2 \sin^{-1} \frac{x}{3}$

Domain $-1 \leq \frac{x}{3} \leq 1$

$-3 \leq x \leq 3$ (1m)

Range $-\frac{\pi}{2} \leq \sin^{-1} \frac{x}{3} \leq \frac{\pi}{2}$

$-\pi \leq y \leq \pi$ (1m)

(e) $\frac{3x}{x-1} \leq 2 \quad x \neq 1$

$3x \frac{(x-1)^2}{x-1} \leq 2(x-1)^2$

$3x(x-1) \leq 2(x-1)^2$

$0 \leq 2(x-1)^2 - 3x(x-1)$

$0 \leq (x-1)(2x-2-3x)$

$0 \leq (-x-2)(x-1)$

$0 \geq (x+2)(x-1)$

$\therefore -2 \leq x \leq 1$

But $x \neq 1 \quad \therefore -2 \leq x < 1$ (3m)

Question 2.

(a) $\frac{d}{dx} x \tan^{-1} x$

$= \tan^{-1} x^2 \cdot 1 + \frac{2x \cdot x}{1+x^4}$

$= \tan^{-1} x^2 + \frac{2x^2}{1+x^4}$ (3m)

(b)(i) $5 \sin x + 3 \cos x$
 $= R(\sin x \cos \alpha + \cos x \sin \alpha)$

$R \cos \alpha = 5 \quad R \sin \alpha = 3$

$\therefore \tan \alpha = \frac{3}{5} \quad \alpha = 31^\circ$ (1m)

$R^2 = 25 + 9 \quad R = \sqrt{34}$ (1m)

(ii) $5 \sin x + 3 \cos x = 4$

$\sqrt{34}(\sin[x+\alpha]) = 4$

$\sin(x+\alpha) = \frac{4}{\sqrt{34}}$

$x+\alpha = 43^\circ, 137^\circ$

$x = 12^\circ, 106^\circ$ (2m)

(c) $(2x - \frac{1}{x^2})^9 = {}^9C_0(2x)^9 + {}^9C_1(2x)^8 \cdot \frac{-1}{x^2} \dots$

required term ${}^9C_3(2x)^6 \left(\frac{-1}{x^2}\right)^3$

$= (-1)^3 {}^9C_3 64$

$= -5376$ (3m)

(d) $I = \int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 - \cos 2x \, dx$

$= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}}$

$= \frac{1}{2} \left[\left(\frac{\pi}{4} - \frac{1}{2} \right) - 0 \right]$

$= \frac{1}{8} (\pi - 2)$ (1m)

Question 3.

(a) (i) $P(2ap, ap^2)$

$x^2 = 4ay$

$y = \frac{4a}{x}$

$\frac{dy}{dx} = \frac{2x}{4a}$

Question 3 cont.

EQUATION OF TANGENT

$y - y_1 = m(x - x_1)$

$y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$ (3m)

(ii) EQUATION OF TANGENT AT Q

$y = qx - aq^2$ (1m)

(iii) $\therefore px - ap^2 = qx - aq^2$

$(p - q)x = ap^2 - aq^2$

$x = a(p - q)(p + q)$

$\therefore x = a(p + q)$

$y = px - ap^2$

$y = ap(p + q) - ap^2$

$y = apq$

$T(a(p + q), apq)$ (2m)

(b) $P(x) \div (x + 1)(x - 2) = Q(x) + \frac{R_x}{(x + 1)(x - 2)}$

$\therefore P(x) = Q(x)(x + 1)(x - 2) + R(x)$

where $R(x) = mx + b$

$P(-1) = Q(-1)(0)(-3) + m(-1) + b$

$3 = -m + b$ (A)

$P(2) = Q(2)(3)(0) + m(2) + b$

$-5 = 2m + b$ (B)

(B) - (A) $-8 = 3m$

$m = -\frac{8}{3}$

FROM (A) $3 = \frac{8}{3} + b \quad \therefore b = \frac{1}{3}$

$R(x) = \frac{1}{3}(-8x + 1)$ (3m)

(C)(i) JOIN M TO E

IN $\triangle MEN$, $\triangle CAN$

$\angle MEN = 90^\circ$ ($\angle$ STANDING ON DIAMETER)

$\angle CAN = 90^\circ$ (GIVEN)

$\angle MNE$ IS COMMON

$\therefore \triangle MEN \parallel \triangle CAN$ (AAA)

$\therefore \angle EMN = \angle NCA$ ($\angle$ OF $\parallel$ LINES)

(ii) $\angle EMN = \angle EDN$ ($\angle$ ON CHORD EN)

$\therefore \angle NCA = \angle EDN$

$\therefore$ EXTERIOR ANGLE = INTERIOR OPPOSITE ANGLE

$\therefore BCED$ IS A CYCLIC QUAD (2m)

Question 4.

(a) (i) $T = A + Ce^{-kt}$

$\frac{dT}{dt} = -k(Ce^{-kt})$

$\frac{dT}{dt} = -k(A + Ce^{-kt} - A)$

$= -k(T - A)$ (1m)

(ii) $T = 20 + Ce^{-kt}$

when $t = 0 \quad T = 500$

$500 = 20 + C$

$C = 480 \quad A = 20^\circ$ (AIR TEMP.) (2m)

$\therefore T = 20 + 480e^{-kt}$

$t = 15 \quad T = 200^\circ$

$200 = 20 + 480e^{-15k}$

$\frac{180}{480} = e^{-15k} \quad (-15k = \ln \frac{180}{480})$

$k = 0.0654$ (1m)

(iii) $T = 20 + 480e^{-0.0654t}$

$100 = 20 + 480e^{-0.0654t}$

$\frac{80}{480} = e^{-0.0654t}$

$t = \frac{-\ln \frac{80}{480}}{0.0654}$ (1m)

$t = 27.4$ minutes $t = 27$ minutes

(b) STEP 1. PROVE TRUE FOR $n = 1$

LHS = $1^3 = 1$ RHS = $\frac{1}{4}(1)^2(2)^2 = 1$

LHS = RHS TRUE FOR $n = 1$

STEP 2. ASSUME TRUE FOR $n = k$

$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}(k^4)(k+1)^2$

PROVE TRUE FOR $n = k + 1$

RHS = $\frac{1}{4}(k+1)^2(k+2)^2$

LHS = $1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3$

$= \frac{1}{4}(k^4)(k+1)^2 + (k+1)^3$

$= \frac{1}{4}(k+1)^2 \{ k^2 + 4(k+1) \}$

$= \frac{1}{4}(k+1)^2 (k^2 + 4k + 4)$

$= \frac{1}{4}(k+1)^2 (k+2)^2$

$=$ RHS. TRUE FOR $n = k + 1$

STEP 3. PROVED TRUE FOR $n = 1$

STEP 2 IMPLIES TRUE FOR $n = 2, 3, 4, \dots$

$\therefore$ BY THE PRINCIPLE OF MATHEMATICAL INDUCTION TRUE FOR ALL n (3m)

Question 4 cont.

$$\begin{aligned} (c) \quad x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= -1 - \frac{f(x_1)}{f'(x_1)} \\ &= -2.67 \quad (2m) \end{aligned}$$

$$\begin{aligned} (d) \quad f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \quad (2m) \end{aligned}$$

Question 5.

$$\begin{aligned} (a) \text{ WALLABY} &= \frac{7!}{2!2!} \leftarrow 2 \text{ A's} \quad \leftarrow 2 \text{ L's} \\ &= 1260. \quad (1m) \end{aligned}$$

$$\begin{aligned} (b) \quad n C_r + n C_{r+1} &= \frac{n!}{(n-r)!r!} + \frac{n!}{(n-r-1)!(r+1)!} \\ &= \frac{n!(r+1)}{(n-r)!(r+1)!} + \frac{n!(n-r)}{(n-r)!(r+1)!} \\ &= \frac{n! + n! - r n!}{(n-r)!(r+1)!} \\ &= \frac{n+1!}{(n-r)!(r+1)!} = {}^{n+1}C_{r+1} \quad (3m) \end{aligned}$$

$$(c) (i) \quad \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} = \frac{125}{216} \quad (1m)$$

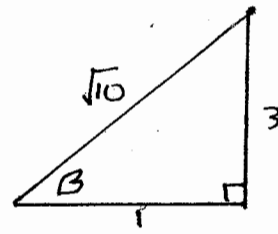
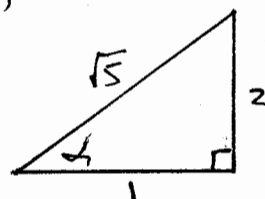
$$(ii) \quad 1 - \frac{125}{216} = \frac{91}{216} \quad (1m)$$

$$\begin{aligned} (iii) \quad \left(\frac{125}{216} + \frac{91}{216}\right)^5 &= {}^5C_0 \left(\frac{125}{216}\right)^5 + \dots \\ &+ {}^5C_3 \left(\frac{125}{216}\right)^2 \left(\frac{91}{216}\right)^3 \end{aligned}$$

$$\therefore P(\text{exactly 3 turns}) = 10 \frac{125^2 \cdot 91^3}{(216)^5} \quad (2m)$$

$$\begin{aligned} (iv) \quad P(\text{AT LEAST ONE 6}) &= 1 - P(\text{NO 6's}) \quad (A)-(B) \\ &= 1 - {}^5C_0 \left(\frac{125}{216}\right)^5 \\ &= 1 - \left(\frac{125}{216}\right)^5 \quad (2m) \end{aligned}$$

(d)



$$\begin{aligned} \text{let } \cos \alpha &= \frac{1}{5} \quad \text{let } \cos \beta = \frac{1}{10} \\ \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \frac{1}{5} \cdot \frac{1}{10} - \frac{2}{5} \cdot \frac{3}{10} \\ &= \frac{-5}{50} = -\frac{1}{10} \end{aligned}$$

$$\begin{aligned} \therefore \cos(\alpha + \beta) &= -\frac{1}{10} \\ \therefore \alpha + \beta &= \frac{3\pi}{4} \\ \therefore \cos^{-1} \frac{1}{5} + \cos^{-1} \frac{1}{10} &= \frac{3\pi}{4} \quad (2m) \end{aligned}$$

Question 6.

$$\begin{aligned} (a) (i) \quad \frac{d^2x}{dt^2} &= -n^2x \\ \frac{d}{dt} \left(\frac{1}{2} v^2 \right) &= -n^2x \\ \frac{dx}{dt} \cdot \frac{1}{2} v &= -n^2 \int x dx \\ &= -\frac{n^2 x^2}{2} + C \end{aligned}$$

$$\text{When } x=a \quad v=0 \quad \therefore C = \frac{n^2 a^2}{2}$$

$$\begin{aligned} \therefore \frac{1}{2} v^2 &= \frac{n^2 a^2}{2} - \frac{n^2 x^2}{2} \\ v^2 &= n^2(a^2 - x^2) \quad (2m) \end{aligned}$$

$$(ii) \text{ Using } v^2 = n^2(a^2 - x^2)$$

$$25 = n^2(a^2 - (0.2)^2) \quad (A)$$

$$9 = n^2(a^2 - (0.6)^2) \quad (B)$$

$$\begin{aligned} (A) \quad \frac{25}{9} &= \frac{a^2 - 0.04}{a^2 - 0.36} \\ (B) \end{aligned}$$

$$25a^2 - 9 = 9a^2 - 0.36$$

$$16a^2 = 8.64$$

$$a^2 = 0.54$$

$$a = 0.735 \quad (1m)$$

$$16 = 0.32n^2$$

$$n^2 = 50$$

$$n = \sqrt{50}$$

$$\text{Frequency} = \frac{n}{2\pi}$$

$$= \frac{\sqrt{50}}{2\pi} \quad (1m)$$

Question 6 cont.

(iii) MAX VELOCITY OCCURS AT $x=0$
 $V^2 = u^2 a^2$

$$V^2 = 27 \quad V = 3\sqrt{3}$$

$$\frac{2}{3} V_{\max} = 2\sqrt{3}$$

$$V^2 = u^2 (a^2 - x^2)$$

$$12 = 50(0.54 - x^2)$$

$$x^2 = 0.3$$

$$x = 0.548 \text{ m}$$

(2m)

(b)(i) $\tan 40^\circ = \frac{h}{OP}$

$$OP = \frac{h}{\tan 40^\circ}$$

Similarly $OQ = \frac{h}{\tan 30^\circ}$ (2m)

(ii) As $\triangle POQ$ IS RIGHT ANGLED

$$OP^2 + OQ^2 = 100^2$$

$$\frac{h^2}{\tan^2 40^\circ} + \frac{h^2}{\tan^2 30^\circ} = 100^2$$

$$\frac{h^2 \tan^2 30^\circ + h^2 \tan^2 40^\circ}{\tan^2 30^\circ \tan^2 40^\circ} = 100^2$$

$$h^2 = 100^2 \frac{(\tan^2 30^\circ + \tan^2 40^\circ)}{\tan^2 30^\circ + \tan^2 40^\circ}$$

$$h = \frac{100(\tan 30^\circ \tan 40^\circ)}{\sqrt{\tan^2 30^\circ + \tan^2 40^\circ}} \quad (2m)$$

(iii) $h = 47.57 \text{ m}$

$$OP = \frac{h}{\tan 40^\circ} = 56.69 \text{ m}$$

$$\therefore \cos \angle OPQ = \frac{OP}{100} = 0.5669$$

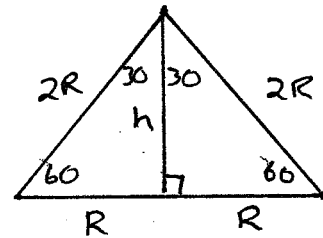
$$\angle OPQ = 55.47^\circ$$

Bearing Q from P = 055°

$\therefore$ Bearing P from Q = 235°
 2m

Question 7.

(a)(i) IF SEMI VERTICAL ANGLE IS 30°



By PYTHAGORAS $h = \sqrt{3}R$

$$V_{\text{cone}} = \frac{1}{3} \pi R^2 h$$

$$= \frac{1}{3} \pi R^2 \sqrt{3}R$$

$$= \frac{\sqrt{3}}{3} \pi R^3 \quad (2m)$$

(ii) $\frac{dv}{dt} = \frac{dv}{dr} \cdot \frac{dr}{dt}$

$$\frac{dv}{dr} = \sqrt{3} \pi r^2 = 4\sqrt{3} \pi \quad (R=2)$$

$$\frac{dv}{dt} = \frac{dv}{dr} \cdot \frac{dr}{dt}$$

$$1 = 4\sqrt{3} \pi \cdot \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{4\sqrt{3} \pi} \text{ m/min} \quad (2m)$$

(b)(i) $f(x) = x^2 - 4x + 5$
 $= (x-2)^2 + 1$

$\therefore$ vertex (2, 1) (1m)

$\therefore$ largest positive domain $x \geq 2$

FOR INVERSE

(ii) Let $x = (y-2)^2 + 1$

$$x-1 = (y-2)^2$$

$$y-2 = \pm \sqrt{x-1}$$

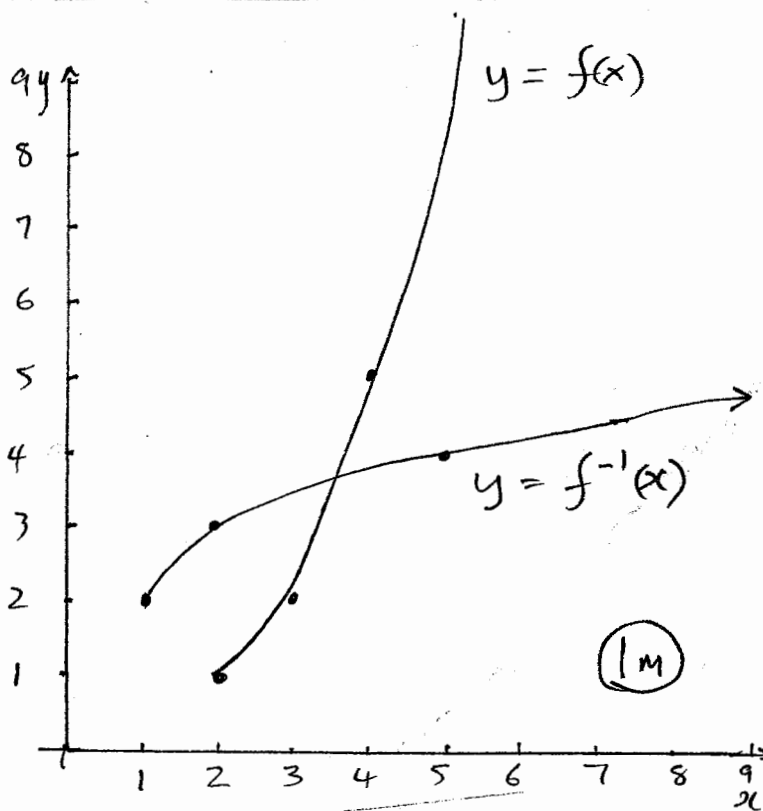
$$y = 2 \pm \sqrt{x-1}$$

$$f^{-1}(x) = 2 + \sqrt{x-1}$$

$$\text{as } y \geq 2$$

$$\therefore f^{-1}(x) \geq 2$$

Question 7 cont.



c) i) For MAX HEIGHT $y = 0$

$$0 = v \sin \theta - gt$$

$$t = \frac{v \sin \theta}{g}$$

sub $y = v \sin \theta \left(\frac{v \sin \theta}{g} \right) - \frac{1}{2} g \left(\frac{v \sin \theta}{g} \right)^2$

(Hm) $y_m = \frac{v^2 \sin^2 \theta}{2g}$ (1m)

ii) from $x = v \cos \theta t$

$$t = \frac{x}{v \cos \theta}$$

$$y = \frac{v \sin \theta x}{v \cos \theta} - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2$$

$$y = \tan \theta x - \frac{g x^2}{2 v^2 \cos^2 \theta}$$

$$y = \tan \theta x - \frac{g x^2 \sec^2 \theta}{2 v^2} \quad (2m)$$

c) iii) Put a, b into eqⁿ of path

$$b = a \tan \theta - \frac{g a^2 \sec^2 \theta}{2 v^2}$$

$$\frac{g a^2 \sec^2 \theta}{2 v^2} = a \tan \theta - b$$

$$\frac{g a^2 \sec^2 \theta}{a \tan \theta - b} = 2 v^2$$

$$v^2 = \frac{g a^2 \sec^2 \theta}{2 (a \tan \theta - b)}$$

subst v^2 into H

$$H = \frac{v^2 \sin^2 \theta}{2g}$$

$$H = \frac{g a^2 \sec^2 \theta \sin^2 \theta}{2g \cdot 2 (a \tan \theta - b)}$$

$$H = \frac{a^2 \tan^2 \theta}{4 (a \tan \theta - b)} \quad (2m)$$